

Comprehensive Study Notes: Diagonalization of Matrices

Welcome to **Diagonalization**! In engineering and computer science (e.g., graphics rendering, population modeling, and Markov chains), computing high powers of a dense matrix A^n can require thousands of multiplication operations.

Diagonalization is a mathematical technique that simplifies a complicated square matrix into a **diagonal matrix**—where numbers only exist on the main diagonal, and everything else is zero. Computing D^n for a diagonal matrix takes milliseconds because you simply raise each diagonal number to the power n !

1. High-Level Overview: Core Concepts

What is Diagonalization?

A square matrix A of size $n \times n$ is **diagonalizable** if it can be factored into three special matrices:

$$A = PDP^{-1} \quad \text{or equivalently} \quad P^{-1}AP = D$$

- A : The original $n \times n$ matrix.
- P (**Modal Matrix**): A matrix whose **columns are the linearly independent eigenvectors** of A .
- D (**Spectral Matrix**): A **diagonal matrix** whose main diagonal entries are the corresponding **eigenvalues** of A .
- P^{-1} : The inverse of matrix P .

Key Theorems to Remember

1. **The Diagonalization Theorem**: An $n \times n$ matrix A is diagonalizable **if and only if** it has n linearly independent eigenvectors.
2. **Distinct Eigenvalues Condition**: If an $n \times n$ matrix A has n **distinct (all different)** eigenvalues, then A is **guaranteed** to be diagonalizable.
3. **Repeated Eigenvalues Condition**: If an eigenvalue repeats, the matrix can *still* be diagonalizable as long as the repeated eigenvalue produces enough linearly independent eigenvectors to total n eigenvectors overall.

2. Step-by-Step Procedure to Diagonalize A

Follow this step-by-step algorithm to diagonalize any matrix on an exam:

1. **Find Eigenvalues (λ)**: Solve the characteristic equation $|A - \lambda I| = 0$.
2. **Find Eigenvectors (X_i)**: Solve $(A - \lambda I)X = 0$ for each eigenvalue to get n linearly independent eigenvectors $X_1, X_2, \dots, X_n$.
3. **Construct Modal Matrix P** : Combine eigenvectors as column vectors:

$$P = [X_1 \quad X_2 \quad \cdots \quad X_n]$$

4. **Construct Spectral Matrix D** : Place eigenvalues on the diagonal matching the exact column order of P :

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

5. **Calculate P^{-1}** : Find the inverse of P using row operations $[P \mid I] \rightarrow [I \mid P^{-1}]$ or formulas.
6. **Write Final Form**: Express $A = PDP^{-1}$ or verify $P^{-1}AP = D$.

3. Power and Inverse of a Matrix using Eigen-Decomposition

Once a matrix is in its diagonalized form $A = PDP^{-1}$, calculating powers and inverses becomes trivial.

Calculating Powers (A^n)

Multiplying A by itself n times causes intermediate $P^{-1}P$ pairs to collapse into identity matrices I :

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD(P^{-1}P)DP^{-1} = PD^2P^{-1}$$

By mathematical induction:

$$A^n = PD^nP^{-1}$$

Where D^n is simply:

$$D^n = \begin{bmatrix} \lambda_1^n & 0 & \cdots & 0 \\ 0 & \lambda_2^n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^n \end{bmatrix}$$

Calculating Inverses (A^{-1})

Similarly, the inverse of matrix A via eigen-decomposition is:

$$A^{-1} = PD^{-1}P^{-1}$$

Where D^{-1} is formed by taking the reciprocal of each diagonal entry:

$$D^{-1} = \begin{bmatrix} \lambda_1^{-1} & 0 & \cdots & 0 \\ 0 & \lambda_2^{-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^{-1} \end{bmatrix}$$

4. Fully Worked Examples

Example 1: 2x2 Matrix Diagonalization & Powers

Problem: Diagonalize the matrix $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ and find A^3 .

Step 1: Find Eigenvalues

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} -\lambda & 1 \\ -2 & -3 - \lambda \end{vmatrix} = 0 \\ (-\lambda)(-3 - \lambda) - (1)(-2) &= 0 \implies \lambda^2 + 3\lambda + 2 = 0 \\ (\lambda + 1)(\lambda + 2) &= 0 \implies \lambda_1 = -1, \quad \lambda_2 = -2 \end{aligned}$$

Step 2: Find Eigenvectors

- For $\lambda_1 = -1$:

$$(A + I)X = 0 \implies \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Row operation $R_2 \rightarrow R_2 + 2R_1$ gives $x + y = 0 \implies y = -x$.

Let $x = k$; then $X_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

- For $\lambda_2 = -2$:

$$(A + 2I)X = 0 \implies \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Row operation $R_2 \rightarrow R_2 + R_1$ gives $2x + y = 0 \implies y = -2x$.

Let $x = k$; then $X_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

Step 3: Construct P , D , and P^{-1}

- **Modal Matrix P :** $P = [X_1 \ X_2] = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$

- **Spectral Matrix D :** $D = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$

- **Find P^{-1} :**

Using Row Reduction on $[P \mid I]$:

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ -1 & -2 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 + R_1} \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & -1 & 1 & 1 \end{array} \right]$$

$$\xrightarrow{R_2 \rightarrow -R_2} \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & -1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{cc|cc} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & -1 \end{array} \right]$$

Thus, $P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$.

Step 4: Diagonalization Matrix Expression

$$A = PDP^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

Step 5: Compute A^3

Using $A^3 = PD^3P^{-1}$:

$$D^3 = \begin{bmatrix} (-1)^3 & 0 \\ 0 & (-2)^3 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -8 \end{bmatrix}$$

$$PD^3 = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -8 \end{bmatrix} = \begin{bmatrix} -1 & -8 \\ 1 & 16 \end{bmatrix}$$

$$A^3 = (PD^3)P^{-1} = \begin{bmatrix} -1 & -8 \\ 1 & 16 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 6 & 7 \\ -14 & -15 \end{bmatrix}$$

Example 2: 3x3 Matrix with Repeated Eigenvalues

Problem: Diagonalize $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$.

Step 1: Find Eigenvalues

Solving $|A - \lambda I| = 0$ yields:

$$-\lambda^3 - 3\lambda^2 + 4 = 0 \implies \lambda = 1, -2, -2$$

Step 2: Find Eigenvectors

- **For Distinct Eigenvalue $\lambda = 1$:**

$$(A - I)X = 0 \implies \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Row reduction to Echelon form gives equations:

$$x_1 + x_2 = 0, \quad x_2 + x_3 = 0$$

Setting free variable $x_2 = k$ gives $x_1 = -k$ and $x_3 = -k$.

$$X_1 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$$

- **For Repeated Eigenvalue $\lambda = -2$:**

$$(A + 2I)X = 0 \implies \begin{bmatrix} 3 & 3 & 3 \\ -3 & -3 & -3 \\ 3 & 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Row reducing yields only **one** independent row equation:

$$x_1 + x_2 + x_3 = 0 \implies x_3 = -x_1 - x_2$$

Because we have **two free variables**, we can generate **two linearly independent eigenvectors**:

Let $x_1 = a$ and $x_2 = b$:

$$X = \begin{bmatrix} a \\ b \\ -a - b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

Thus:

$$X_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

Step 3: Construct P and D

Since we obtained 3 linearly independent eigenvectors in total, A is **diagonalizable**!

- **Modal Matrix P :** $P = [X_1 \quad X_2 \quad X_3] = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ -1 & -1 & -1 \end{bmatrix}$

- **Spectral Matrix D :** $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$

Step 4: Inverse P^{-1} & Final Decomposition

By row reducing $[P \mid I] \rightarrow [I \mid P^{-1}]$, we get:

$$P^{-1} = \begin{bmatrix} -1 & -1 & -1 \\ 0 & -1 & -1 \\ 1 & 2 & 1 \end{bmatrix}$$

Final Diagonalized Representation ($A = PDP^{-1}$):

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 \\ 0 & -1 & -1 \\ 1 & 2 & 1 \end{bmatrix}$$